



FINAL REPORT

Tourism Experience Analytics

Using AI and machine learning to deliver personalized travel experiences and predictive business insights for tourism stakeholders.

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Project Scope and AI Capabilities

An end-to-end system integrating regression, classification, and collaborative filtering for travel platforms.

Rating Prediction

Utilizes **regression models** to forecast potential attraction ratings before a trip or campaign launch.

Visit Mode Classification

Categorizes visitors into segments like **Family, Solo, or Couples** for highly targeted marketing offers.

Attraction Recommender

Employs **collaborative filtering** to suggest attractions based on historical user patterns and similarities.



Solving the Personalization Challenge

Addressing the industry's need for scalable user behavior understanding and retention strategies.

Predictive Accuracy
Moving beyond reactive feedback by
predicting satisfaction before the
journey begins.

Discovery Alignment
Matching users with attractions that
align with their **documented history**
and preferences.



Hyper-Segmentation
Identifying the **unique needs** of
business, solo, and family travelers at
scale.

DATASET OVERVIEW

Scale of the Integrated Dataset

A robust repository of tourism transactions spanning nearly a decade of global travel activity.



52,930

Total Transactions

Full volume of recorded visits and reviews analyzed from 2013 to 2022.

33,530

Unique Users

Scale of individual traveler profiles within the collaborative filtering matrix.

30

Attractions Analyzed

Core points of interest tracked across various continents and regions.

4.158

Average User Rating

Baseline sentiment across the entire dataset indicating high general satisfaction.

Automated Data Cleaning Pipeline

Ensuring data integrity and modeling readiness through a reproducible Python-based workflow.



Location Enrichment

Filling missing categorical labels like **CityName** and **Country** with 'Unknown' labels.

AUTO

Consistency Enforcement

Standardizing **numeric ID columns** and temporal fields for model consumption.



Rating Validation

Clipping and validating ratings to ensure all values fall within the strict **1-5 range**.



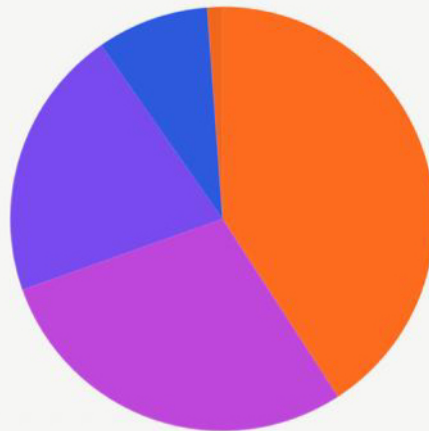
Final Transformation

Exporting the processed set of **52,930 rows and 21 features** for ML training.

EDA ANALYSIS

Visitor Segment Distribution

Analyzing how different travel cohorts contribute to overall platform transaction volume.



40.85% Couples 28.75% Family 20.68% Friends 8.55% Solo
1.18% Business

Geographic Market Concentration

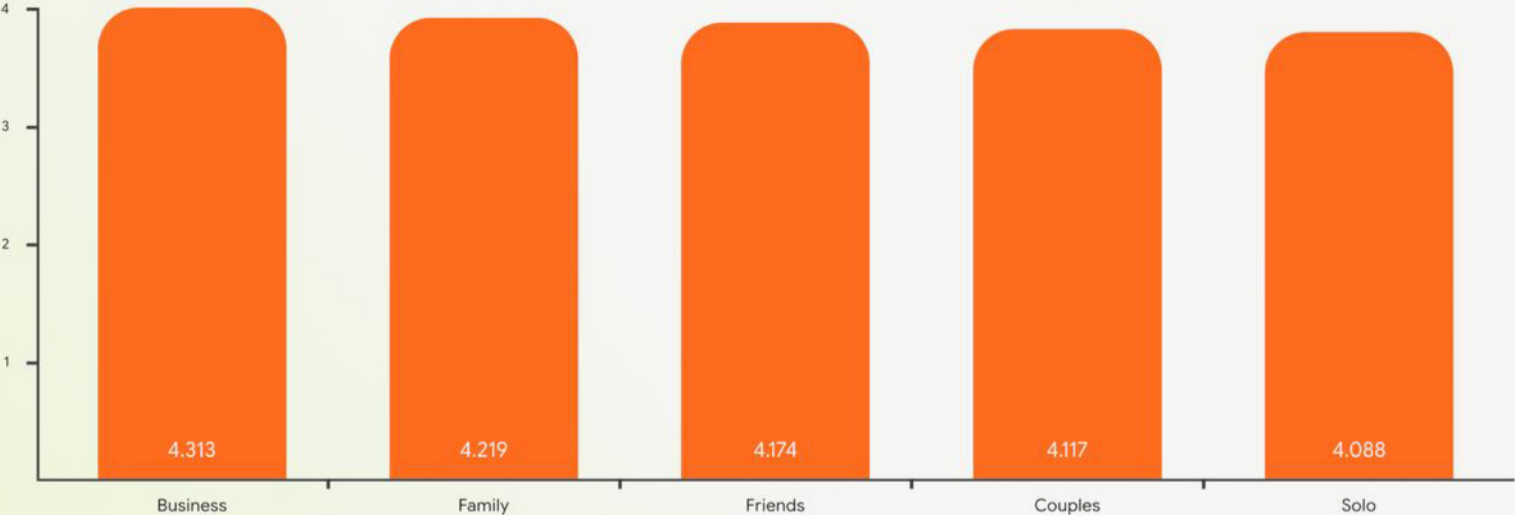
Identifying the primary source markets driving attraction visits and platform usage.

Rank	Country	Transactions
1	Australia	13,322
2	United Kingdom	6,722
3	United States	6,261
4	Indonesia	4,842
5	Singapore	2,807

VISITOR INSIGHTS

Sentiment by Visit Category

How traveler context influences the average satisfaction levels and ratings.



Predictive Modeling Strategy

Selection and evaluation of top-performing algorithms for regression and classification.



Regression Architecture

Utilized **GradientBoostingRegressor** as the top model Evaluated using MAE and RMSE on hold-out splits Targets directional satisfaction forecasting



Classification Framework

Deployed **RandomForestClassifier** for segment ID Applied leakage prevention for VisitModel Optimized for weighted F1-Score performance



Recommendation Engine

User-based collaborative filtering via **KNN**
Calculated using **cosine similarity** in a matrix
Generates top-N unseen attraction suggestions

Model Performance Evaluation

Transparent assessment of model accuracy and real-world predictive utility.

Task	Metric	Result
Regression	RMSE	0.9136
Regression	MAE	0.7165
Classification	Accuracy	49.30%
Classification	F1 (Weighted)	0.4411

BUSINESS INSIGHTS

Strategic Recommendations

Data-driven actions to improve user experience, targeting, and platform retention.



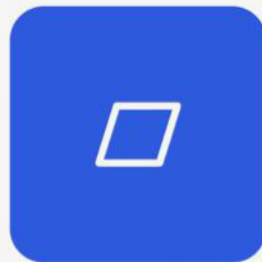
Prioritize **Couples and Family-focused** campaign bundles as these segments dominate volume.



Use **rating prediction** to proactively identify and flag potentially low-satisfaction attraction contexts.



Personalize **package design and messaging** based on the predicted visit mode classification.



Capture **richer context features** like trip duration and budget to improve recommendation hit rates.

STREAMLIT APP

Interactive Analytics Platform

A deployable Streamlit application designed for business stakeholders and executive oversight.



01

Executive Dashboard: Real-time visualization of visitor distributions and top-performing attractions.



02

Prediction Forms: Interactive tools for forecasting ratings and classifying traveler modes on the fly.



03

Recommendation Engine: Live UI to generate personalized attraction suggestions for specific user IDs.



04

Performance Panel: Dedicated space for model evaluation metrics to ensure system transparency.

FINAL REVIEW

Project Conclusion & Roadmap

Summarizing achievements in AI-driven tourism analytics and outlining the roadmap for future model and data improvements.

